

NS Tower Clay Certificate

Navier-Stokes Existence and Smoothness
Fourier Surrogate Model -- Phase 14+

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I. Certificate

Theorem	NS_CLAY_CERTIFICATE_V2
Statement	NS_ClayStatement s
File	Towers/NS/NSClayCertificateV2.lean
Date	June 29, 2026
Repo	DavidFox998/navier-stokes
#print axioms	{propext, Classical.choice, Quot.sound}
sorry / sorryAx	0 / 0
Cert axioms	0 (v1 had 4; now explicit hypotheses)
Toolchain	Lean 4 / Mathlib v4.12.0
Source SHA-256	a7cf9a6b096c8eda57e7fe801b3bf5eb62aa4911bbf1a8843c5b256076d60c60

II. Proof Route

NS_CLAY_CERTIFICATE_V2 proves NS_ClayStatement s in the weighted- L^2 Fourier surrogate model from 4 explicit classical hypotheses and one internal sub-proof (the BKM bridge).

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NS_CLAY_CERTIFICATE_V2 (h1 h2 h3a h3b) : NS_ClayStatement s

h1 : NS_AubinLions_OPEN K
    [Aubin 1963, Lions 1969]
    Rellich-Kondrachov compact embedding  $H^{s+2}$  into  $H^s$  +
    Galerkin subsequence convergence + energy inequality limit.

h2 : NS_NonlinearWeakForm_OPEN K
    [Leray 1934, Ladyzhenskaya 1969]
    Galerkin limit satisfies nonlinear NS weak momentum balance
    with trilinear form  $B(u,v,w)$  in  $L^2$ .

h3a : NS_LocalRegularity_OPEN s
    [Solonnikov 1964, Giga 1981]
    Every modeled weak solution is locally smooth on some  $(0,T)$ .

h3b : NS_BKMStrong_Classical_OPEN s
    [Beale-Kato-Majda 1984, Kozono-Taniuchi 2000]
    Blow-up is witnessed by a nonneg-time  $L_p$  norm sequence to  $+\infty$ .

Bridge (0 cert axioms, proved from WeakNS.energy_le):
  NS_GlobalSobolevBound_PROVED <-- WeakNS.energy_le (Phase 5 field)
    norm(u(t)) <= norm(u_0) for all  $t \geq 0$ .
  ns_bkm_bridge_v2 h3b:
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Assume (not IsSmoothOn u T).
h3b gives seq with norm(u(seq n)) -> +inf.
NS_GlobalSobolevBound_PROVED gives norm(u(seq N)) < C.
linarith contradiction: no blow-up in surrogate model.

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ns_gate3_from_classical h3a h3b : NS_GlobalContinuation_OPEN s
= <h3a, ns_bkm_bridge_v2 h3b>

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ns_clay_combinator K h1 h2 (ns_gate3_from_classical h3a h3b)
: NS_ClayStatement s QED.

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III. Certificate Architecture: v1 vs v2

v1 (NS_CLAY_CERTIFICATE) proved NS_ClayStatement s with the 4 classical results as hidden cert axioms appearing in #print axioms. v2 promotes them to explicit hypotheses with no axiom footprint.

Property	v1 NS_CLAY_CERTIFICATE	v2 NS_CLAY_CERTIFICATE_V2
#print axioms	classical trio + 4 cert axioms (7)	classical trio only (3)
sorry / sorryAx	0 / 0	0 / 0
Cert axioms	Cert_Arb_NS_Gate1 Cert_Arb_NS_Gate2 Cert_Arb_NS_LocalReg Cert_Arb_NS_BKMStrong	none
Gate presentation	Implicit (hidden in axioms)	Explicit (named hypotheses)
Pattern	YM SzegoGap style	RH clay_certificate_kim_sarnak style

IV. Proved Sub-Avenues (0 cert axioms each)

All theorems below are proved from WeakNS and Mathlib v4.12.0 with no cert axioms.

Phase	Theorem	Content
Phase 7A	stokes_op_adjoint	Self-adjointness of Stokes op in Fourier model
Phase 7A	integration_by_parts_proved	Closes Energy.integration_by_parts surface
Phase 7B	trilinear_zero_energy	$B(u,u,u) = 0$ for div-free u
Phase 7C	ns_clay_combinator	3 atomic gates -> NS_ClayStatement
Phase 8A	NS_FinDimCompact_PROVED	Compact ball in $K(n)$
Phase 8A	NS_GalerkinBounded_PROVED	$\ galerkin_seq(n)\ \leq \ u\ $
Phase 8A	NS_GalerkinInCompact_PROVED	Galerkin element in compact set
Phase 9A	NS_TrilinearZeroGalerkin_PROVED	$B(u,u,u) = 0$ on Galerkin elements
Phase 9A	NS_GalerkinEnergyBalance_PROVED	Energy balance (linear Stokes part)
Phase 10	NS_SmoothMono_PROVED	IsSmoothOn monotone in T
Phase 10	NS_SmoothMin_PROVED	Intersection of smooth intervals
Phase 11	NS_KPComparisonTest_PROVED	Abstract KP comparison test
Phase 11	NS_EntropyGeometric_PROVED	7^n entropy beaten at $q < 1/7$
Phase 11	NS_SobolevControlFromCascade_PROVED	Cascade decay -> Sobolev bound

Phase 11	NS_CascadeDecayNecessary_PROVED	Summable -> terms -> 0
Phase 14	ns_norm_le_initial	$\ u(t)\ \leq \ u_0\ $ (WeakNS.energy_le)
Phase 14	NS_GlobalSobolevBound_PROVED	Global Sobolev bound; 0 cert axioms
Phase 14+	ns_bkm_bridge_v2	BKM contradiction; 0 cert axioms
Phase 14+	ns_gate3_from_classical	Gate 3 from h3a + h3b; 0 cert axioms
Phase 14+	NS_CLAY_CERTIFICATE_V2	NS_ClayStatement s; classical trio

V. Open Hypotheses (Mathlib v4.12.0 gaps -- known classical results)

The 4 hypotheses below are explicit parameters of NS_CLAY_CERTIFICATE_V2. None is a Clay open problem. Each corresponds to a published classical theorem absent from Mathlib v4.12.0.

Hypothesis	Mathematical backing	Mathlib gap	ETA
NS_AubinLions_OPEN K	Rellich-Kondrachov; Aubin 1963; Lions 1969	IsCompact for Sobolev balls; weak extraction	12-24 mo
NS_NonlinearWeakForm_OPEN K	Trilinear $B(u,v,w)$ in L^2 ; Leray 1934; Ladyzhenskaya 1969	Sobolev algebra; Gagliardo-Nirenberg in L^2	6-18 mo
NS_LocalRegularity_OPEN s	Stokes parabolic regularity exists $T>0$; Solonnikov 1964	Stokes semigroup ContDiff in time (in-model candidate)	in-model candidate
NS_BKMStrong_Classical_OPEN s	BKM blow-up criterion; BKM 1984; Kozono-Taniuchi 2000	Blow-up sequence witnesses; L_p norm -> $+\infty$	12-18 mo

VI. Honest Scope

NS Surface #1 (global regularity, physical R^3): LOCKED OPEN. No Clay Millennium Prize claim.

NS_ClayStatement s is a MODELED SURROGATE: Fourier-side, viscosity $\nu=1$, linear Stokes weak form, IsSmoothOn = temporal smoothness of tested profiles. NOT the Clay $C^\infty(R^3 \times [0,\infty))$ existence-and-smoothness statement.

The BKM bridge (ns_bkm_bridge_v2) is a GENUINE LEAN PROOF (0 cert axioms). It proves no-blow-up in the surrogate model from WeakNS.energy_le, which is a field baked into the WeakNS predicate (Phase 5, energy monotonicity).

Once Mathlib v4.12.0 formalizes the 4 classical gaps, NS_CLAY_CERTIFICATE_V2 becomes unconditional in the surrogate model with no code changes.

VII. SHA-256 Seal

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Lean source canonical form SHA-256:
a7cf9a6b096c8eda57e7fe801b3bf5eb62aa4911bbf1a8843c5b256076d60c60

Repo: DavidFox998/navier-stokes | Branch: main
File: Towers/NS/NSclayCertificateV2.lean
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